# A PROPER DOMINANT MORPHISM WITH FINITE FUNCTION FIELD EXTENSION IS FINITE OVER A NEIGHBOURHOOD OF A CODIMENSION ONE POINT

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

*Allowed standard prerequisites (editorial).*

- Properness with finite fibres in a neighbourhood, and the algebraic criterion in codimension one. Cohomology of Schemes, label coherent-lemma-proper-finite-fibre-finite-in-neighbourhood (proof line 2543); Algebra, label algebra-lemma-finite-in-codim-1 (proof line 2553). The first reduces the statement to the finiteness of the fibre over the point, and the second provides the algebraic criterion giving finiteness over a neighbourhood in the one-dimensional case.

*Source reference.* Original lemma 02MA: sources/stacks-chow/tags.txt, source lines 3437–3437. Navigation pointer for \ref{algebra-lemma-finite-in-codim-1} used inside the counted span (line 2553); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 02OH: sources/stacks-chow/tags.txt, source lines 3516–3516. Navigation pointer for \ref{coherent-lemma-proper-finite-fibre-finite-in-neighbourhood} used inside the counted span (line 2543); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 0AB7: sources/stacks-chow/tags.txt, source lines 13010–13010. Navigation pointer for \ref{varieties-lemma-finite-in-codim-1} used inside the counted span (line 2540); the argument that consumes it is retained in the span.

---

*Date*: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $f : X \to Y$ be a morphism of schemes. Let $\xi \in Y$ be a point. Assume that*

(1) *$X$, $Y$ are integral,*
(2) *$Y$ is locally Noetherian*
(3) *$f$ is proper, dominant and $R(Y) \subset R(X)$ is finite, and*
(4) *$\dim(\mathcal{O}_{Y,\xi}) = 1$.*

*Then there exists an open neighbourhood $V \subset Y$ of $\xi$ such that $f|_{f^{-1}(V)} : f^{-1}(V) \to V$ is finite.*

*Proof.* This lemma is a special case of Varieties, Lemma 0AB7. Here is a direct argument in this case. By Cohomology of Schemes, Lemma 02OH it suffices to prove that $f^{-1}(\{\xi\})$ is finite. We replace $Y$ by an affine open, say $Y = \mathrm{Spec}(R)$. Note that $R$ is Noetherian, as $Y$ is assumed locally Noetherian. Since $f$ is proper it is quasi-compact. Hence we can find a finite affine open covering $X = U_1 \cup \ldots \cup U_n$ with each $U_i = \mathrm{Spec}(A_i)$. Note that $R \to A_i$ is a finite type injective homomorphism of domains such that the induced extension of fraction fields is finite. Thus the lemma follows from Algebra, Lemma 02MA. □